

A Scattered-Orbit Partial Result for Invariant Strictly Convex Renormings

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Source Question

The source paper is Michal Doucha, *Invariant strictly convex renormings*, arXiv:2508.16182v1. Question 7.1 asks:

Do there exist a separable Banach space X and a surjective linear isometry $T : X \rightarrow X$ such that there is no T -invariant strictly convex renorming of X ?

Equivalently, this is the cyclic, or $\mathbb{Z}$ -action, case of the invariant strict convex renorming problem. Question 7.2 asks the corresponding question for countable amenable groups.

Result

This packet proves a partial positive result for induced actions on $C(K)$. It does not solve Questions 7.1 or 7.2 in full, but it removes a natural source of possible cyclic counterexamples.

Theorem 1. *Let G be a group acting by homeomorphisms on a compact metrizable space K . Assume that, for every $p \in K$, the orbit closure*

$$K_p = \overline{G \cdot p}$$

is scattered. Then the induced isometric action of G on $C(K)$ admits an equivalent G -invariant strictly convex norm.

Corollary 1. *For every homeomorphism $h : [0, 1] \rightarrow [0, 1]$, the composition isometry*

$$Tf = f \circ h \quad (f \in C[0, 1])$$

admits an equivalent T -invariant strictly convex norm on $C[0, 1]$.

Idea of the Proof

The key observation is that an invariant strictly convex renorming can be built from countably many equivariant maps into already strictly convex G -spaces, provided those maps separate points. For $C(K)$, a nonzero function is detected on some orbit closure K_p . If K_p is scattered and metrizable, then $C(K_p)$ is separable and has the Radon–Nikodym property. Lancien’s invariant LUR renorming

theorem gives a G -invariant locally uniformly rotund norm on $C(K_p)$, hence a strictly convex one. The restriction maps

$$C(K) \rightarrow C(K_p)$$

therefore supply enough equivariant strictly convex coordinates to separate all nonzero functions in $C(K)$.

Proof

Lemma 1. *Let X be a separable Banach space and let G act on X by linear isometries. Suppose that for every nonzero $x \in X$ there are a Banach G -space Y_x , an equivalent G -invariant strictly convex norm on Y_x , and a bounded G -equivariant linear map*

$$A_x : X \rightarrow Y_x$$

such that $A_x x \neq 0$. Then X admits an equivalent G -invariant strictly convex norm.

Proof. For each nonzero $x \in X$, fix such a map A_x and write $|\cdot|_x$ for the chosen invariant strictly convex norm on Y_x . The sets

$$U_x = \{z \in X : |A_x z|_x > |A_x x|_x / 2\}$$

form an open cover of $X \setminus \{0\}$. Since X is separable, choose a countable subcover U_{x_n} , $n \in \mathbb{N}$. After multiplying each A_{x_n} by a positive scalar, assume $\|A_{x_n}\| \leq 1$, where the range norm is $|\cdot|_{x_n}$.

Let

$$Y = \left(\bigoplus_{n=1}^{\infty} Y_{x_n} \right)_{\ell_2}$$

with its diagonal G -action and its ℓ_2 -sum norm. This norm is G -invariant and strictly convex. Define

$$A : X \rightarrow Y, \quad Az = (2^{-n} A_{x_n} z)_{n=1}^{\infty}.$$

The map A is bounded and G -equivariant. It is injective: if $z \neq 0$, then $z \in U_{x_n}$ for some n , so $A_{x_n} z \neq 0$. Therefore

$$\|z\|' = \|z\|_X + \|Az\|_Y$$

is an equivalent G -invariant strictly convex norm on X . Indeed, the first summand is an equivalent invariant norm and the second summand is a strictly convex norm on X because A is injective. $\square$

Proof of Theorem 1. Let $0 \neq f \in C(K)$. Pick $p \in K$ with $f(p) \neq 0$. The orbit closure $K_p = \overline{G \cdot p}$ is compact, metrizable, scattered, and G -invariant. The restriction map

$$R_p : C(K) \rightarrow C(K_p), \quad R_p(g) = g|_{K_p},$$

is bounded, linear, and G -equivariant, and $R_p f \neq 0$.

Since K_p is compact metrizable, $C(K_p)$ is separable. Since K_p is scattered, the classical $C(K)$ -criterion gives that $C(K_p)$ has the Radon–Nikodym property. Lancien’s theorem says that every separable G -Banach space with the Radon–Nikodym property admits an equivalent G -invariant locally uniformly rotund norm. In particular, $C(K_p)$ admits an equivalent G -invariant strictly convex norm.

Thus every nonzero $f \in C(K)$ is detected by a bounded equivariant map into a G -space with an invariant strictly convex norm. Lemma 1 applies and gives the desired invariant strictly convex renorming of $C(K)$. $\square$

Proof of Corollary 1. If h is increasing and $p \in [0, 1]$, then either p is fixed, or p lies in a component (a, b) of $[0, 1] \setminus \text{Fix}(h)$. On such a component all iterates move monotonically toward one endpoint in positive time and toward the other endpoint in negative time. Hence

$$\overline{\{h^n(p) : n \in \mathbb{Z}\}}$$

is either a singleton or a countable compact set with at most two accumulation points. It is scattered.

If h is decreasing, apply the same argument to the increasing homeomorphism h^2 . The h -orbit closure of p is the union of at most two h^2 -orbit closures, hence is again compact, countable, and scattered. Theorem 1 now applies with $G = \mathbb{Z}$. $\square$

Verification and Limitations

The argument is purely structural; no computation is used. The main external inputs are standard: $C(L)$ has the Radon–Nikodym property when L is compact scattered, and Lancien’s invariant LUR theorem for separable RNP G -spaces.

The theorem does not cover orbit closures containing perfect minimal parts, such as adding-machine components in the generic Cantor homeomorphism. It therefore does not settle the $C(2^{\mathbb{N}})$ generic-action question from Doucha’s Question 7.4, nor the full cyclic Question 7.1.

Novelty Check

The run indexes were searched for arXiv:2508.16182, invariant strictly convex renormings, cyclic/ $\mathbb{Z}$ -actions, and $C(K)$ variants. Web searches were also made for exact phrases including “T-invariant strictly convex renorming”, “Z-invariant strictly convex renorming”, “cyclic group invariant strictly convex renorming”, and related AFGR/Lancien terminology. No exact prior answer to Questions 7.1–7.2 or to the scattered orbit-closure subcase was found. Because the proof is a short combination of known tools, it should be treated as a likely-valid partial observation rather than a high novelty full solution.

References

- M. Doucha, *Invariant strictly convex renormings*, arXiv:2508.16182v1, 2025.
- G. Lancien, *Dentability indices and locally uniformly convex renormings*, Rocky Mountain J. Math. 23 (1993), 635–647.
- R. Deville, G. Godefroy, and V. Zizler, *Smoothness and Renormings in Banach Spaces*, Pitman Monographs and Surveys in Pure and Applied Mathematics 64, 1993.

Human Review Recommendation

Check the separating-family lemma and the use of the $C(K)$ -RNP criterion. If those standard inputs are accepted, the packet gives a clean positive subcase: single composition isometries on $C[0, 1]$ cannot witness a negative answer to Doucha’s cyclic question.